

Generation Interconnection System Impact Study Report

For Queue Project Q23-2283

State Street 100 kV

May 27, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2283. The Interconnection Customer proposes a 99.5 MW solar photovoltaic facility interconnecting at the State Street 100 kV substation in the Indiana area. The request was received on January 26, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 636 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,278,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: STATE ST - W OLYMPIA 100 kV
Contingency: WEST OLYMPIA - STATE ST 100 kV Cir. 2
Loading: 174.9% (Rating: 167 MVA)
DFAX: 5.8%
Pre-project loading: 171.5%
2. Facility: STATE ST - WEST OLYMPIA 100 kV Line
Contingency: STATE STREET - WEST OLYMPIA 100 kV
Loading: 174.9% (Rating: 167 MVA)
DFAX: 5.8%
Pre-project loading: 171.5%
3. Facility: W OLYMPIA - STATE STREET 100 kV
Contingency: STATE ST - W OLYMPIA 100 kV L
Loading: 174.9% (Rating: 167 MVA)
DFAX: 5.8%
Pre-project loading: 171.5%
4. Facility: HARPER - MONTE DEL ESTADO 6303 100 kV
Contingency: RICHLAND SUBSTATION - DEKALB 500 kV
Loading: 105.6% (Rating: 532 MVA)
DFAX: 6.1%
Pre-project loading: 104.5%
5. Facility: MONROEVILLE - WOODBURN 100 kV
Contingency: RICHLAND - DEKALB 500 kV
Loading: 103.0% (Rating: 398 MVA)
DFAX: 5.1%
Pre-project loading: 101.7%
6. Facility: MINNEHAHA - STATION S VAN NUYS 230 kV
Contingency: RICHLAND - TROY ENERGY LLC 500 kV Line
Loading: 102.9% (Rating: 1,890 MVA)

DFAX: 10.1%

Pre-project loading: 102.4%

7. Facility: BARRET - ROSEAU COUNTY 100 kV

Contingency: HARPER - MONTE DEL ESTADO 6303 SUBSTATION 100 kV

Loading: 102.8% (Rating: 444 MVA)

DFAX: 5.4%

Pre-project loading: 101.6%

8. Facility: WOODBURN - SAGE 100 kV line

Contingency: RICHLAND - DEKALB 500 kV

Loading: 102.4% (Rating: 413 MVA)

DFAX: 5.1%

Pre-project loading: 101.1%

9. Facility: MONTE DEL ESTADO 6303 - SAGE 100 kV

Contingency: RICHLAND - DEKALB 500 kV

Loading: 101.6% (Rating: 434 MVA)

DFAX: 5.1%

Pre-project loading: 100.4%

10. Facility: DEAN FOREST - MUSKOGEE 230 kV Line

Contingency: MUSKOGEE - SWAN VLY 500 kV Line

Loading: 100.1% (Rating: 1,145 MVA)

DFAX: 8.3%

Pre-project loading: 99.4%

11. Facility: EASTOWN RD - DEAN FOREST 230 kV

Contingency: SWAN VALLEY - MUSKOGEE 500 kV

Loading: 100.1% (Rating: 1,128 MVA)

DFAX: 8.3%

Pre-project loading: 99.4%

12. Facility: MUSKOGEE - DARLINGTON RD SUBSTATION 100 kV L

Contingency: DEKALB - RICHLAND 500 kV

Loading: 100.1% (Rating: 783 MVA)

DFAX: 9.0%

Pre-project loading: 98.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W OLYMPIA - STATE ST 100 kV Cir. 1	Rebuild line with higher-capacity conductor	\$1,726,000	23% (\$395,000)
W OLYMPIA - STATE STREET 100 kV #2	Rebuild line with higher-capacity conductor	\$1,693,000	23% (\$387,000)
WEST OLYMPIA - STATE ST 100 kV Cir. 3	Rebuild line with higher-capacity conductor	\$1,778,000	23% (\$406,000)
HARPER - MONTE DEL ESTADO 6303 100 kV	Reconductor line	\$1,491,000	4% (\$54,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MONROEVILLE - WOODBURN 100 kV	Reconductor line	\$2,973,000	22% (\$646,000)
MINNEHAHA - STATION S VAN NUYS 230 kV	Reconductor line	\$4,795,000	1% (\$36,000)
BARRET - ROSEAU COUNTY 100 kV	Reconductor line	\$3,216,000	16% (\$508,000)
SAGE - WOODBURN 100 kV	Reconductor line	\$966,000	22% (\$210,000)
MONTE DEL ESTADO 6303 - SAGE 100 kV	Reconductor line	\$1,173,000	22% (\$255,000)
DEAN FOREST - MUSKOGEE 230 kV	Reconductor line	\$14,508,000	1% (\$167,000)
EASTOWN RD - DEAN FOREST 230 kV L	Reconductor line	\$11,044,000	1% (\$127,000)
MUSKOGEE - DARLINGTON RD 100 kV	Reconductor line	\$6,080,000	1% (\$87,000)
	Total allocated to project		\$3,278,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.